Huawei HCCDA-AI

5-Phase • 100 MCQ Exam Preparation

FOURTH EDITION

100 Fresh Exam-Practice Questions

Edition: This is a separate 4th-edition question bank. It follows the uploaded HCCDA-AI course scope while using fresh question angles and wording rather than copying the previous edition documents.

Distribution: AI 20 • ML 20 • DL 20 • CV 20 • Huawei Cloud + DevOps/MLOps 20.

Format: Question → 4 Options → Correct Answer → Why Correct → Why Every Other Option Is Incorrect → Final Answer Key.

Phase 1 — Artificial Intelligence (AI)

20 questions in this phase.

Question 1: What does the term 'intelligent behavior' in AI most closely refer to?

- **A.** The ability of a system to perform tasks that normally require forms of human intelligence
- **B.** The ability to increase hard-disk capacity
- **C.** The ability to change monitor brightness
- **D.** The ability to install an operating system

Correct Answer: A. The ability of a system to perform tasks that normally require forms of human intelligence

Why this is correct: The ability of a system to perform tasks that normally require forms of human intelligence directly matches the concept tested.

Why B is incorrect: The ability to increase hard-disk capacity does not correctly answer the question.

Why C is incorrect: The ability to change monitor brightness does not correctly answer the question.

Why D is incorrect: The ability to install an operating system does not correctly answer the question.

Question 2: Which example best represents a narrow AI application?

- **A.** A system specialized in recognizing objects in images
- **B.** A hypothetical machine capable of every intellectual task
- **C.** A file backup utility
- **D.** A keyboard driver

Correct Answer: A. A system specialized in recognizing objects in images

Why this is correct: A system specialized in recognizing objects in images directly matches the concept tested.

Why B is incorrect: A hypothetical machine capable of every intellectual task does not correctly answer the question.

Why C is incorrect: A file backup utility does not correctly answer the question.

Why D is incorrect: A keyboard driver does not correctly answer the question.

Question 3: What is one purpose of studying AI history?

- **A.** To understand how major AI approaches and capabilities evolved
- **B.** To determine the size of a cloud bucket
- **C.** To replace current AI systems
- **D.** To configure a network switch

Correct Answer: A. To understand how major AI approaches and capabilities evolved

Why this is correct: To understand how major AI approaches and capabilities evolved directly matches the concept tested.

Why B is incorrect: To determine the size of a cloud bucket does not correctly answer the question.

Why C is incorrect: To replace current AI systems does not correctly answer the question.

Why D is incorrect: To configure a network switch does not correctly answer the question.

Question 4: Which approach is most associated with logical rules and explicitly encoded knowledge?

A. Symbolic AI

B. Batch processing

C. Object storage

D. Image interpolation

Correct Answer: A. Symbolic AI

Why this is correct: Symbolic AI directly matches the concept tested.

Why B is incorrect: Batch processing does not correctly answer the question.

Why C is incorrect: Object storage does not correctly answer the question.

Why D is incorrect: Image interpolation does not correctly answer the question.

Question 5: Which area is included among the AI domains in the course?

A. Robotics

B. Spreadsheet formatting

C. Printer maintenance

D. Disk defragmentation

Correct Answer: A. Robotics

Why this is correct: Robotics directly matches the concept tested.

Why B is incorrect: Spreadsheet formatting does not correctly answer the question.

Why C is incorrect: Printer maintenance does not correctly answer the question.

Why D is incorrect: Disk defragmentation does not correctly answer the question.

Question 6: Why can data availability become an AI adoption challenge?

A. A useful AI system may require sufficient relevant data for development and evaluation

B. AI never uses data

C. Data only affects screen resolution

D. Data determines keyboard layout

Correct Answer: A. A useful AI system may require sufficient relevant data for development and evaluation

Why this is correct: A useful AI system may require sufficient relevant data for development and evaluation directly matches the concept tested.

Why B is incorrect: AI never uses data does not correctly answer the question.

Why C is incorrect: Data only affects screen resolution does not correctly answer the question.

Why D is incorrect: Data determines keyboard layout does not correctly answer the question.

Question 7: Why can regulation influence an AI project's design?

A. Legal and governance requirements can affect how data and AI systems are used

B. Regulation controls CPU clock speed

C. Regulation replaces model training

D. Regulation is only about file names

Correct Answer: A. Legal and governance requirements can affect how data and AI systems are used

Why this is correct: Legal and governance requirements can affect how data and AI systems are used directly matches the concept tested.

Why B is incorrect: Regulation controls CPU clock speed does not correctly answer the question.

Why C is incorrect: Regulation replaces model training does not correctly answer the question.

Why D is incorrect: Regulation is only about file names does not correctly answer the question.

Question 8: What is meant by an AI industry ecosystem?

A. The network of technologies, organizations, services, applications, and stakeholders around AI

B. A single computer's file system

C. A camera sensor

D. A Python loop

Correct Answer: A. The network of technologies, organizations, services, applications, and stakeholders around AI

Why this is correct: The network of technologies, organizations, services, applications, and stakeholders around AI directly matches the concept tested.

Why B is incorrect: A single computer's file system does not correctly answer the question.

Why C is incorrect: A camera sensor does not correctly answer the question.

Why D is incorrect: A Python loop does not correctly answer the question.

Question 9: Why are current AI trends relevant to AI professionals?

A. They help professionals understand changing technologies, applications, and skills

B. They determine monitor size

C. They replace fundamental concepts

D. They eliminate practical work

Correct Answer: A. They help professionals understand changing technologies, applications, and skills

Why this is correct: They help professionals understand changing technologies, applications, and skills directly matches the concept tested.

Why B is incorrect: They determine monitor size does not correctly answer the question.

Why C is incorrect: They replace fundamental concepts does not correctly answer the question.

Why D is incorrect: They eliminate practical work does not correctly answer the question.

Question 10: Which combination is explicitly represented in Huawei Cloud EI capabilities in the course?

A. Vision, NLP, speech, and OCR

B. Payroll, accounting, and word processing

C. BIOS, drivers, and disk repair

D. Only database replication

Correct Answer: A. Vision, NLP, speech, and OCR

Why this is correct: Vision, NLP, speech, and OCR directly matches the concept tested.

Why B is incorrect: Payroll, accounting, and word processing does not correctly answer the question.

Why C is incorrect: BIOS, drivers, and disk repair does not correctly answer the question.

Why D is incorrect: Only database replication does not correctly answer the question.

Question 11: What is the main purpose of a requirements document?

A. To record what the proposed system should accomplish and the constraints it must satisfy

B. To store model weights

C. To replace testing

D. To select computer hardware

Correct Answer: A. To record what the proposed system should accomplish and the constraints it must satisfy

Why this is correct: To record what the proposed system should accomplish and the constraints it must satisfy directly matches the concept tested.

Why B is incorrect: To store model weights does not correctly answer the question.

Why C is incorrect: To replace testing does not correctly answer the question.

Why D is incorrect: To select computer hardware does not correctly answer the question.

Question 12: Why are stakeholder interviews useful during AI requirements work?

A. They can reveal user needs, constraints, workflows, and expectations

B. They train the model automatically

C. They create OCR output

D. They replace deployment

Correct Answer: A. They can reveal user needs, constraints, workflows, and expectations

Why this is correct: They can reveal user needs, constraints, workflows, and expectations directly matches the concept tested.

Why B is incorrect: They train the model automatically does not correctly answer the question.

Why C is incorrect: They create OCR output does not correctly answer the question.

Why D is incorrect: They replace deployment does not correctly answer the question.

Question 13: Which item is most likely a functional requirement?

A. The application shall classify uploaded images into defined categories

B. The application should be highly scalable

C. The service should have low latency

D. The system should be secure

Correct Answer: A. The application shall classify uploaded images into defined categories

Why this is correct: The application shall classify uploaded images into defined categories directly matches the concept tested.

Why B is incorrect: The application should be highly scalable does not correctly answer the question.

Why C is incorrect: The service should have low latency does not correctly answer the question.

Why D is incorrect: The system should be secure does not correctly answer the question.

Question 14: Which item is most likely a non-functional requirement?

A. The prediction API should respond within an agreed latency target

B. The application shall extract text from an invoice

C. The system shall detect people

D. The application shall generate a report

Correct Answer: A. The prediction API should respond within an agreed latency target

Why this is correct: The prediction API should respond within an agreed latency target directly matches the concept tested.

Why B is incorrect: The application shall extract text from an invoice does not correctly answer the question.

Why C is incorrect: The system shall detect people does not correctly answer the question.

Why D is incorrect: The application shall generate a report does not correctly answer the question.

Question 15: Why should project scope be defined before implementation?

A. It establishes boundaries and keeps development aligned with the intended objective

B. It guarantees model accuracy

C. It eliminates all risks

D. It chooses the programming language automatically

Correct Answer: A. It establishes boundaries and keeps development aligned with the intended objective

Why this is correct: It establishes boundaries and keeps development aligned with the intended objective directly matches the concept tested.

Why B is incorrect: It guarantees model accuracy does not correctly answer the question.

Why C is incorrect: It eliminates all risks does not correctly answer the question.

Why D is incorrect: It chooses the programming language automatically does not correctly answer the question.

Question 16: What is a user goal in an AI project?

A. A desired outcome that the user expects the system to help achieve

B. A model weight

C. A cloud region

D. A camera pixel

Correct Answer: A. A desired outcome that the user expects the system to help achieve

Why this is correct: A desired outcome that the user expects the system to help achieve directly matches the concept tested.

Why B is incorrect: A model weight does not correctly answer the question.

Why C is incorrect: A cloud region does not correctly answer the question.

Why D is incorrect: A camera pixel does not correctly answer the question.

Question 17: Why might an AI solution use a hybrid edge-cloud design?

A. Different parts of processing can be placed where latency, compute, connectivity, and privacy requirements fit best

B. Hybrid systems never use networks

C. Hybrid means two labels per image

D. Hybrid systems cannot use cloud services

Correct Answer: A. Different parts of processing can be placed where latency, compute, connectivity, and privacy requirements fit best

Why this is correct: Different parts of processing can be placed where latency, compute, connectivity, and privacy requirements fit best directly matches the concept tested.

Why B is incorrect: Hybrid systems never use networks does not correctly answer the question.

Why C is incorrect: Hybrid means two labels per image does not correctly answer the question.

Why D is incorrect: Hybrid systems cannot use cloud services does not correctly answer the question.

Question 18: What should an architecture diagram primarily communicate?

A. Major components, interfaces, data flow, and relationships in the solution

B. Only source-code indentation

C. Only the project logo

D. Only exam scores

Correct Answer: A. Major components, interfaces, data flow, and relationships in the solution

Why this is correct: Major components, interfaces, data flow, and relationships in the solution directly matches the concept tested.

Why B is incorrect: Only source-code indentation does not correctly answer the question.

Why C is incorrect: Only the project logo does not correctly answer the question.

Why D is incorrect: Only exam scores does not correctly answer the question.

Question 19: Why is exam-domain review useful before a certification test?

A. It helps organize revision around the knowledge areas that may be assessed

B. It reveals the exact future questions

C. It replaces all learning

D. It guarantees a passing score

Correct Answer: A. It helps organize revision around the knowledge areas that may be assessed

Why this is correct: It helps organize revision around the knowledge areas that may be assessed directly matches the concept tested.

Why B is incorrect: It reveals the exact future questions does not correctly answer the question.

Why C is incorrect: It replaces all learning does not correctly answer the question.

Why D is incorrect: It guarantees a passing score does not correctly answer the question.

Question 20: What is the purpose of a certification practice quiz?

A. To test understanding and expose topics that need further revision

B. To change cloud credentials

C. To train a production model

D. To replace the official certification

Correct Answer: A. To test understanding and expose topics that need further revision

Why this is correct: To test understanding and expose topics that need further revision directly matches the concept tested.

Why B is incorrect: To change cloud credentials does not correctly answer the question.

Why C is incorrect: To train a production model does not correctly answer the question.

Why D is incorrect: To replace the official certification does not correctly answer the question.

Phase 2 — Machine Learning (ML)

20 questions in this phase.

Question 1: What is an input feature in a supervised ML dataset?

A. A measurable variable supplied to a model as input

B. The final exam result

C. A cloud access key

D. A camera driver

Correct Answer: A. A measurable variable supplied to a model as input

Why this is correct: A measurable variable supplied to a model as input directly matches the concept tested.

Why B is incorrect: The final exam result does not correctly answer the question.

Why C is incorrect: A cloud access key does not correctly answer the question.

Why D is incorrect: A camera driver does not correctly answer the question.

Question 2: What is a label in a classification dataset?

A. The known class associated with a training example

B. The dataset's file extension

C. The cloud region

D. The model's learning rate

Correct Answer: A. The known class associated with a training example

Why this is correct: The known class associated with a training example directly matches the concept tested.

Why B is incorrect: The dataset's file extension does not correctly answer the question.

Why C is incorrect: The cloud region does not correctly answer the question.

Why D is incorrect: The model's learning rate does not correctly answer the question.

Question 3: What is the main purpose of splitting data into training and evaluation subsets?

A. To assess whether learned patterns generalize beyond the examples used for learning

B. To increase CPU frequency

C. To remove all missing values automatically

D. To change the class definitions

Correct Answer: A. To assess whether learned patterns generalize beyond the examples used for learning

Why this is correct: To assess whether learned patterns generalize beyond the examples used for learning directly matches the concept tested.

Why B is incorrect: To increase CPU frequency does not correctly answer the question.

Why C is incorrect: To remove all missing values automatically does not correctly answer the question.

Why D is incorrect: To change the class definitions does not correctly answer the question.

Question 4: Why should preprocessing steps be applied consistently at inference time?

A. The model expects new inputs to be represented in a compatible way with training inputs

B. It guarantees correct predictions

C. It creates new labels

D. It removes the need for validation

Correct Answer: A. The model expects new inputs to be represented in a compatible way with training inputs

Why this is correct: The model expects new inputs to be represented in a compatible way with training inputs directly matches the concept tested.

Why B is incorrect: It guarantees correct predictions does not correctly answer the question.

Why C is incorrect: It creates new labels does not correctly answer the question.

Why D is incorrect: It removes the need for validation does not correctly answer the question.

Question 5: What is imputation?

A. Replacing missing values using a defined strategy

B. Deleting every feature

C. Changing model architecture

D. Uploading data to a notebook

Correct Answer: A. Replacing missing values using a defined strategy

Why this is correct: Replacing missing values using a defined strategy directly matches the concept tested.

Why B is incorrect: Deleting every feature does not correctly answer the question.

Why C is incorrect: Changing model architecture does not correctly answer the question.

Why D is incorrect: Uploading data to a notebook does not correctly answer the question.

Question 6: Why can irrelevant features be harmful?

A. They may add noise or unnecessary complexity without useful predictive information

B. They always increase accuracy

C. They are required by every algorithm

D. They change the cloud provider

Correct Answer: A. They may add noise or unnecessary complexity without useful predictive information

Why this is correct: They may add noise or unnecessary complexity without useful predictive information directly matches the concept tested.

Why B is incorrect: They always increase accuracy does not correctly answer the question.

Why C is incorrect: They are required by every algorithm does not correctly answer the question.

Why D is incorrect: They change the cloud provider does not correctly answer the question.

Question 7: What is a classification decision boundary?

A. A boundary in feature space separating regions assigned to different classes

B. A physical server boundary

C. A storage bucket

D. An API endpoint

Correct Answer: A. A boundary in feature space separating regions assigned to different classes

Why this is correct: A boundary in feature space separating regions assigned to different classes directly matches the concept tested.

Why B is incorrect: A physical server boundary does not correctly answer the question.

Why C is incorrect: A storage bucket does not correctly answer the question.

Why D is incorrect: An API endpoint does not correctly answer the question.

Question 8: What is the purpose of a baseline metric?

A. It provides a reference level against which later model improvements can be compared

B. It guarantees production performance

C. It replaces the test set

D. It defines cloud pricing

Correct Answer: A. It provides a reference level against which later model improvements can be compared

Why this is correct: It provides a reference level against which later model improvements can be compared directly matches the concept tested.

Why B is incorrect: It guarantees production performance does not correctly answer the question.

Why C is incorrect: It replaces the test set does not correctly answer the question.

Why D is incorrect: It defines cloud pricing does not correctly answer the question.

Question 9: What does a high recall with lower precision generally indicate?

A. The model finds many actual positives but also accepts more false positives

B. The model misses nearly every positive

C. The model has no predictions

D. The model has zero errors

Correct Answer: A. The model finds many actual positives but also accepts more false positives

Why this is correct: The model finds many actual positives but also accepts more false positives directly matches the concept tested.

Why B is incorrect: The model misses nearly every positive does not correctly answer the question.

Why C is incorrect: The model has no predictions does not correctly answer the question.

Why D is incorrect: The model has zero errors does not correctly answer the question.

Question 10: What does a high precision with lower recall generally indicate?

A. Positive predictions are often correct, but some actual positives are missed

B. Every positive is detected

C. The model has no false positives and no false negatives

D. The model cannot classify anything

Correct Answer: A. Positive predictions are often correct, but some actual positives are missed

Why this is correct: Positive predictions are often correct, but some actual positives are missed directly matches the concept tested.

Why B is incorrect: Every positive is detected does not correctly answer the question.

Why C is incorrect: The model has no false positives and no false negatives does not correctly answer the question.

Why D is incorrect: The model cannot classify anything does not correctly answer the question.

Question 11: Why can class imbalance distort plain accuracy?

A. A dominant class can make overall accuracy look good even when minority-class performance is poor

B. Accuracy ignores all predictions

C. Imbalance affects only hardware

D. Accuracy is always invalid

Correct Answer: A. A dominant class can make overall accuracy look good even when minority-class performance is poor

Why this is correct: A dominant class can make overall accuracy look good even when minority-class performance is poor directly matches the concept tested.

Why B is incorrect: Accuracy ignores all predictions does not correctly answer the question.

Why C is incorrect: Imbalance affects only hardware does not correctly answer the question.

Why D is incorrect: Accuracy is always invalid does not correctly answer the question.

Question 12: What is stratified sampling useful for in classification?

A. Preserving approximate class proportions when creating subsets

B. Changing image resolution

C. Creating cloud APIs

D. Replacing labels with random values

Correct Answer: A. Preserving approximate class proportions when creating subsets

Why this is correct: Preserving approximate class proportions when creating subsets directly matches the concept tested.

Why B is incorrect: Changing image resolution does not correctly answer the question.

Why C is incorrect: Creating cloud APIs does not correctly answer the question.

Why D is incorrect: Replacing labels with random values does not correctly answer the question.

Question 13: What is an ML experiment configuration?

A. The collection of settings used for a particular training run

B. A cloud invoice

C. A camera frame

D. A PDF heading

Correct Answer: A. The collection of settings used for a particular training run

Why this is correct: The collection of settings used for a particular training run directly matches the concept tested.

Why B is incorrect: A cloud invoice does not correctly answer the question.

Why C is incorrect: A camera frame does not correctly answer the question.

Why D is incorrect: A PDF heading does not correctly answer the question.

Question 14: Why should experiment results be recorded?

A. They allow different runs to be compared and successful configurations to be traced

B. They remove the need for data

C. They guarantee deployment

D. They change the model automatically

Correct Answer: A. They allow different runs to be compared and successful configurations to be traced

Why this is correct: They allow different runs to be compared and successful configurations to be traced directly matches the concept tested.

Why B is incorrect: They remove the need for data does not correctly answer the question.

Why C is incorrect: They guarantee deployment does not correctly answer the question.

Why D is incorrect: They change the model automatically does not correctly answer the question.

Question 15: What is a performance regression?

A. A decrease in relevant system or model performance after a change

B. An increase in storage

C. A new dataset label

D. A successful API call

Correct Answer: A. A decrease in relevant system or model performance after a change

Why this is correct: A decrease in relevant system or model performance after a change directly matches the concept tested.

Why B is incorrect: An increase in storage does not correctly answer the question.

Why C is incorrect: A new dataset label does not correctly answer the question.

Why D is incorrect: A successful API call does not correctly answer the question.

Question 16: Why compare models on the same evaluation protocol?

A. A consistent protocol makes performance comparisons more meaningful

B. It guarantees identical models

C. It removes all errors

D. It changes the dataset

Correct Answer: A. A consistent protocol makes performance comparisons more meaningful

Why this is correct: A consistent protocol makes performance comparisons more meaningful directly matches the concept tested.

Why B is incorrect: It guarantees identical models does not correctly answer the question.

Why C is incorrect: It removes all errors does not correctly answer the question.

Why D is incorrect: It changes the dataset does not correctly answer the question.

Question 17: What is a prediction score?

A. A numerical value produced by a model that can support a prediction decision

B. A cloud account number

C. A file size

D. A monitor refresh rate

Correct Answer: A. A numerical value produced by a model that can support a prediction decision

Why this is correct: A numerical value produced by a model that can support a prediction decision directly matches the concept tested.

Why B is incorrect: A cloud account number does not correctly answer the question.

Why C is incorrect: A file size does not correctly answer the question.

Why D is incorrect: A monitor refresh rate does not correctly answer the question.

Question 18: Why can changing a classification threshold alter precision and recall?

A. It changes which scores are accepted as positive predictions

B. It changes the training dataset automatically

C. It changes the model architecture

D. It changes image resolution

Correct Answer: A. It changes which scores are accepted as positive predictions

Why this is correct: It changes which scores are accepted as positive predictions directly matches the concept tested.

Why B is incorrect: It changes the training dataset automatically does not correctly answer the question.

Why C is incorrect: It changes the model architecture does not correctly answer the question.

Why D is incorrect: It changes image resolution does not correctly answer the question.

Question 19: What is concept drift?

A. A change in the relationship between inputs and the target that can make an old model less effective

B. A change in disk size

C. A change in file extension

D. A change in monitor brightness

Correct Answer: A. A change in the relationship between inputs and the target that can make an old model less effective

Why this is correct: A change in the relationship between inputs and the target that can make an old model less effective directly matches the concept tested.

Why B is incorrect: A change in disk size does not correctly answer the question.

Why C is incorrect: A change in file extension does not correctly answer the question.

Why D is incorrect: A change in monitor brightness does not correctly answer the question.

Question 20: Why is periodic model evaluation important after deployment?

A. Real-world data and conditions can change after a model is released

B. Production models never change

C. Evaluation deletes the model

D. It only measures storage capacity

Correct Answer: A. Real-world data and conditions can change after a model is released

Why this is correct: Real-world data and conditions can change after a model is released directly matches the concept tested.

Why B is incorrect: Production models never change does not correctly answer the question.

Why C is incorrect: Evaluation deletes the model does not correctly answer the question.

Why D is incorrect: It only measures storage capacity does not correctly answer the question.

Phase 3 — Deep Learning (DL)

20 questions in this phase.

Question 1: What is a neuron activation?

A. The output produced by applying an activation function to a neuron's computed input

B. A cloud bucket

C. A dataset filename

D. A camera coordinate

Correct Answer: A. The output produced by applying an activation function to a neuron's computed input

Why this is correct: The output produced by applying an activation function to a neuron's computed input directly matches the concept tested.

Why B is incorrect: A cloud bucket does not correctly answer the question.

Why C is incorrect: A dataset filename does not correctly answer the question.

Why D is incorrect: A camera coordinate does not correctly answer the question.

Question 2: Why does a deep network need nonlinear activations?

A. Without nonlinearity, stacking linear layers would still behave like a single linear transformation

B. They increase disk space

C. They create labels

D. They replace training data

Correct Answer: A. Without nonlinearity, stacking linear layers would still behave like a single linear transformation

Why this is correct: Without nonlinearity, stacking linear layers would still behave like a single linear transformation directly matches the concept tested.

Why B is incorrect: They increase disk space does not correctly answer the question.

Why C is incorrect: They create labels does not correctly answer the question.

Why D is incorrect: They replace training data does not correctly answer the question.

Question 3: What does a neural network loss value represent?

A. How far the model's predictions are from the desired targets according to the chosen objective

B. The number of cloud users

C. The number of image pixels

D. The size of a Python file

Correct Answer: A. How far the model's predictions are from the desired targets according to the chosen objective

Why this is correct: How far the model's predictions are from the desired targets according to the chosen objective directly matches the concept tested.

Why B is incorrect: The number of cloud users does not correctly answer the question.

Why C is incorrect: The number of image pixels does not correctly answer the question.

Why D is incorrect: The size of a Python file does not correctly answer the question.

Question 4: What is gradient descent?

A. An optimization approach that updates parameters in a direction intended to reduce the objective

B. An image resizing method

C. A storage protocol

D. A camera calibration technique

Correct Answer: A. An optimization approach that updates parameters in a direction intended to reduce the objective

Why this is correct: An optimization approach that updates parameters in a direction intended to reduce the objective directly matches the concept tested.

Why B is incorrect: An image resizing method does not correctly answer the question.

Why C is incorrect: A storage protocol does not correctly answer the question.

Why D is incorrect: A camera calibration technique does not correctly answer the question.

Question 5: What is an optimizer state?

A. Additional information maintained by some optimizers to determine parameter updates

B. A cloud bucket name

C. A class label

D. An image histogram

Correct Answer: A. Additional information maintained by some optimizers to determine parameter updates

Why this is correct: Additional information maintained by some optimizers to determine parameter updates directly matches the concept tested.

Why B is incorrect: A cloud bucket name does not correctly answer the question.

Why C is incorrect: A class label does not correctly answer the question.

Why D is incorrect: An image histogram does not correctly answer the question.

Question 6: Why are mini-batches commonly used in neural-network training?

A. They provide manageable groups of samples for efficient parameter updates

B. They eliminate the need for data

C. They guarantee convergence

D. They replace the test set

Correct Answer: A. They provide manageable groups of samples for efficient parameter updates

Why this is correct: They provide manageable groups of samples for efficient parameter updates directly matches the concept tested.

Why B is incorrect: They eliminate the need for data does not correctly answer the question.

Why C is incorrect: They guarantee convergence does not correctly answer the question.

Why D is incorrect: They replace the test set does not correctly answer the question.

Question 7: What is the effect of increasing batch size?

A. It changes how many samples contribute to each training update

B. It changes the number of classes automatically

C. It converts CNNs to RNNs

D. It removes all gradients

Correct Answer: A. It changes how many samples contribute to each training update

Why this is correct: It changes how many samples contribute to each training update directly matches the concept tested.

Why B is incorrect: It changes the number of classes automatically does not correctly answer the question.

Why C is incorrect: It converts CNNs to RNNs does not correctly answer the question.

Why D is incorrect: It removes all gradients does not correctly answer the question.

Question 8: Why can early stopping help training?

A. It can stop training when validation performance stops improving, helping limit overfitting

B. It guarantees perfect accuracy

C. It deletes the model

D. It changes image labels

Correct Answer: A. It can stop training when validation performance stops improving, helping limit overfitting

Why this is correct: It can stop training when validation performance stops improving, helping limit overfitting directly matches the concept tested.

Why B is incorrect: It guarantees perfect accuracy does not correctly answer the question.

Why C is incorrect: It deletes the model does not correctly answer the question.

Why D is incorrect: It changes image labels does not correctly answer the question.

Question 9: What is regularization in deep learning?

A. A family of methods intended to improve generalization by discouraging overly complex solutions

B. A cloud deployment method

C. A camera setting

D. A file format

Correct Answer: A. A family of methods intended to improve generalization by discouraging overly complex solutions

Why this is correct: A family of methods intended to improve generalization by discouraging overly complex solutions directly matches the concept tested.

Why B is incorrect: A cloud deployment method does not correctly answer the question.

Why C is incorrect: A camera setting does not correctly answer the question.

Why D is incorrect: A file format does not correctly answer the question.

Question 10: Why can data augmentation help a vision model?

A. It exposes the model to varied versions of training examples and can improve robustness

B. It guarantees no errors

C. It replaces validation

D. It removes the need for labels

Correct Answer: A. It exposes the model to varied versions of training examples and can improve robustness

Why this is correct: It exposes the model to varied versions of training examples and can improve robustness directly matches the concept tested.

Why B is incorrect: It guarantees no errors does not correctly answer the question.

Why C is incorrect: It replaces validation does not correctly answer the question.

Why D is incorrect: It removes the need for labels does not correctly answer the question.

Question 11: What is a convolution's receptive field influenced by?

A. The architecture and preceding operations that determine how much input context affects a feature

B. Only the filename

C. Only the cloud region

D. Only the monitor size

Correct Answer: A. The architecture and preceding operations that determine how much input context affects a feature

Why this is correct: The architecture and preceding operations that determine how much input context affects a feature directly matches the concept tested.

Why B is incorrect: Only the filename does not correctly answer the question.

Why C is incorrect: Only the cloud region does not correctly answer the question.

Why D is incorrect: Only the monitor size does not correctly answer the question.

Question 12: What does stride greater than one generally do in convolution?

A. It moves the filter in larger steps and can reduce spatial output dimensions

B. It always increases resolution

C. It removes channels

D. It changes class labels

Correct Answer: A. It moves the filter in larger steps and can reduce spatial output dimensions

Why this is correct: It moves the filter in larger steps and can reduce spatial output dimensions directly matches the concept tested.

Why B is incorrect: It always increases resolution does not correctly answer the question.

Why C is incorrect: It removes channels does not correctly answer the question.

Why D is incorrect: It changes class labels does not correctly answer the question.

Question 13: Why is padding used in CNNs?

A. It can control border information and the spatial size of convolution outputs

B. It creates cloud credentials

C. It performs OCR

D. It replaces pooling in every model

Correct Answer: A. It can control border information and the spatial size of convolution outputs

Why this is correct: It can control border information and the spatial size of convolution outputs directly matches the concept tested.

Why B is incorrect: It creates cloud credentials does not correctly answer the question.

Why C is incorrect: It performs OCR does not correctly answer the question.

Why D is incorrect: It replaces pooling in every model does not correctly answer the question.

Question 14: What is an encoder-decoder neural architecture often used for?

A. Transforming an input representation into an output representation, including tasks such as segmentation

B. Only storing datasets

C. Only monitoring APIs

D. Only creating invoices

Correct Answer: A. Transforming an input representation into an output representation, including tasks such as segmentation

Why this is correct: Transforming an input representation into an output representation, including tasks such as segmentation directly matches the concept tested.

Why B is incorrect: Only storing datasets does not correctly answer the question.

Why C is incorrect: Only monitoring APIs does not correctly answer the question.

Why D is incorrect: Only creating invoices does not correctly answer the question.

Question 15: Why are skip connections useful in architectures such as U-Net?

A. They provide access to earlier spatial information during decoding

B. They remove the need for training

C. They convert images into audio

D. They replace annotations

Correct Answer: A. They provide access to earlier spatial information during decoding

Why this is correct: They provide access to earlier spatial information during decoding directly matches the concept tested.

Why B is incorrect: They remove the need for training does not correctly answer the question.

Why C is incorrect: They convert images into audio does not correctly answer the question.

Why D is incorrect: They replace annotations does not correctly answer the question.

Question 16: What is transfer learning especially useful for when labeled data is limited?

A. Starting from representations learned on a larger related dataset

B. Eliminating evaluation

C. Guaranteeing perfect accuracy

D. Removing all preprocessing

Correct Answer: A. Starting from representations learned on a larger related dataset

Why this is correct: Starting from representations learned on a larger related dataset directly matches the concept tested.

Why B is incorrect: Eliminating evaluation does not correctly answer the question.

Why C is incorrect: Guaranteeing perfect accuracy does not correctly answer the question.

Why D is incorrect: Removing all preprocessing does not correctly answer the question.

Question 17: What is fine-tuning a pretrained network?

A. Updating some or all pretrained parameters using data for a target task

B. Changing only the filename

C. Deleting pretrained weights

D. Changing cloud storage

Correct Answer: A. Updating some or all pretrained parameters using data for a target task

Why this is correct: Updating some or all pretrained parameters using data for a target task directly matches the concept tested.

Why B is incorrect: Changing only the filename does not correctly answer the question.

Why C is incorrect: Deleting pretrained weights does not correctly answer the question.

Why D is incorrect: Changing cloud storage does not correctly answer the question.

Question 18: Why inspect both training and validation loss?

A. Their relationship can reveal underfitting, overfitting, or training problems

B. It changes GPU hardware

C. It replaces testing

D. It creates API endpoints

Correct Answer: A. Their relationship can reveal underfitting, overfitting, or training problems

Why this is correct: Their relationship can reveal underfitting, overfitting, or training problems directly matches the concept tested.

Why B is incorrect: It changes GPU hardware does not correctly answer the question.

Why C is incorrect: It replaces testing does not correctly answer the question.

Why D is incorrect: It creates API endpoints does not correctly answer the question.

Question 19: Why save the best validation checkpoint rather than only the final epoch?

A. The best validation point may generalize better than a later overfit state

B. The final epoch is always wrong

C. Checkpoints are required for OCR

D. It changes the dataset

Correct Answer: A. The best validation point may generalize better than a later overfit state

Why this is correct: The best validation point may generalize better than a later overfit state directly matches the concept tested.

Why B is incorrect: The final epoch is always wrong does not correctly answer the question.

Why C is incorrect: Checkpoints are required for OCR does not correctly answer the question.

Why D is incorrect: It changes the dataset does not correctly answer the question.

Question 20: What is an output layer's role in a neural network?

A. It converts the learned representation into the form required for the task's prediction

B. It stores cloud credentials

C. It labels the training files

D. It controls monitor brightness

Correct Answer: A. It converts the learned representation into the form required for the task's prediction

Why this is correct: It converts the learned representation into the form required for the task's prediction directly matches the concept tested.

Why B is incorrect: It stores cloud credentials does not correctly answer the question.

Why C is incorrect: It labels the training files does not correctly answer the question.

Why D is incorrect: It controls monitor brightness does not correctly answer the question.

Phase 4 — Computer Vision (CV)

20 questions in this phase.

Question 1: What is spatial resolution in an image?

A. The number of pixels representing its spatial dimensions

B. The number of cloud services

C. The number of object classes

D. The number of API calls

Correct Answer: A. The number of pixels representing its spatial dimensions

Why this is correct: The number of pixels representing its spatial dimensions directly matches the concept tested.

Why B is incorrect: The number of cloud services does not correctly answer the question.

Why C is incorrect: The number of object classes does not correctly answer the question.

Why D is incorrect: The number of API calls does not correctly answer the question.

Question 2: Why can aspect ratio matter when resizing an image?

A. Changing it can distort object geometry and visual content

B. It only affects cloud storage

C. It determines the model's optimizer

D. It changes image labels automatically

Correct Answer: A. Changing it can distort object geometry and visual content

Why this is correct: Changing it can distort object geometry and visual content directly matches the concept tested.

Why B is incorrect: It only affects cloud storage does not correctly answer the question.

Why C is incorrect: It determines the model's optimizer does not correctly answer the question.

Why D is incorrect: It changes image labels automatically does not correctly answer the question.

Question 3: What is cropping?

A. Selecting a spatial region of an image and discarding the rest

B. Changing RGB to audio

C. Creating a cloud endpoint

D. Training an RNN

Correct Answer: A. Selecting a spatial region of an image and discarding the rest

Why this is correct: Selecting a spatial region of an image and discarding the rest directly matches the concept tested.

Why B is incorrect: Changing RGB to audio does not correctly answer the question.

Why C is incorrect: Creating a cloud endpoint does not correctly answer the question.

Why D is incorrect: Training an RNN does not correctly answer the question.

Question 4: Why can image normalization use dataset-specific statistics?

A. Models can be trained with expected channel means and scales that should be matched at inference

B. It creates object annotations

C. It replaces resizing

D. It guarantees accuracy

Correct Answer: A. Models can be trained with expected channel means and scales that should be matched at inference

Why this is correct: Models can be trained with expected channel means and scales that should be matched at inference directly matches the concept tested.

Why B is incorrect: It creates object annotations does not correctly answer the question.

Why C is incorrect: It replaces resizing does not correctly answer the question.

Why D is incorrect: It guarantees accuracy does not correctly answer the question.

Question 5: What is a color channel?

A. A component representing one aspect of color information in an image representation

B. A cloud API

C. A database table

D. A camera password

Correct Answer: A. A component representing one aspect of color information in an image representation

Why this is correct: A component representing one aspect of color information in an image representation directly matches the concept tested.

Why B is incorrect: A cloud API does not correctly answer the question.

Why C is incorrect: A database table does not correctly answer the question.

Why D is incorrect: A camera password does not correctly answer the question.

Question 6: Why can thresholding be useful for document preprocessing?

A. It can separate foreground content from background under suitable conditions

B. It always performs OCR

C. It creates cloud storage

D. It detects faces automatically

Correct Answer: A. It can separate foreground content from background under suitable conditions

Why this is correct: It can separate foreground content from background under suitable conditions directly matches the concept tested.

Why B is incorrect: It always performs OCR does not correctly answer the question.

Why C is incorrect: It creates cloud storage does not correctly answer the question.

Why D is incorrect: It detects faces automatically does not correctly answer the question.

Question 7: What is adaptive thresholding different from a single global threshold?

A. The threshold can vary according to local image conditions

B. It uses no pixels

C. It requires a cloud API

D. It only works on RGB metadata

Correct Answer: A. The threshold can vary according to local image conditions

Why this is correct: The threshold can vary according to local image conditions directly matches the concept tested.

Why B is incorrect: It uses no pixels does not correctly answer the question.

Why C is incorrect: It requires a cloud API does not correctly answer the question.

Why D is incorrect: It only works on RGB metadata does not correctly answer the question.

Question 8: What is erosion useful for in a binary image?

A. It can remove small foreground regions or shrink boundaries

B. It always enlarges objects

C. It creates labels

D. It performs classification

Correct Answer: A. It can remove small foreground regions or shrink boundaries

Why this is correct: It can remove small foreground regions or shrink boundaries directly matches the concept tested.

Why B is incorrect: It always enlarges objects does not correctly answer the question.

Why C is incorrect: It creates labels does not correctly answer the question.

Why D is incorrect: It performs classification does not correctly answer the question.

Question 9: What is dilation useful for?

A. It can expand foreground regions or help close small gaps

B. It always shrinks objects

C. It changes image format only

D. It performs OCR

Correct Answer: A. It can expand foreground regions or help close small gaps

Why this is correct: It can expand foreground regions or help close small gaps directly matches the concept tested.

Why B is incorrect: It always shrinks objects does not correctly answer the question.

Why C is incorrect: It changes image format only does not correctly answer the question.

Why D is incorrect: It performs OCR does not correctly answer the question.

Question 10: What is an image mask?

A. A per-pixel representation used to select or identify a region

B. A cloud credential

C. A neural-network optimizer

D. A video codec

Correct Answer: A. A per-pixel representation used to select or identify a region

Why this is correct: A per-pixel representation used to select or identify a region directly matches the concept tested.

Why B is incorrect: A cloud credential does not correctly answer the question.

Why C is incorrect: A neural-network optimizer does not correctly answer the question.

Why D is incorrect: A video codec does not correctly answer the question.

Question 11: What is a region of interest (ROI)?

A. A selected portion of an image or frame that is relevant to processing

B. A model checkpoint

C. A cloud bucket

D. A Python environment

Correct Answer: A. A selected portion of an image or frame that is relevant to processing

Why this is correct: A selected portion of an image or frame that is relevant to processing directly matches the concept tested.

Why B is incorrect: A model checkpoint does not correctly answer the question.

Why C is incorrect: A cloud bucket does not correctly answer the question.

Why D is incorrect: A Python environment does not correctly answer the question.

Question 12: Why can ROI processing improve an application?

A. It can focus computation on relevant areas and reduce unnecessary processing

B. It always improves model accuracy

C. It removes the need for labels

D. It increases every image's resolution

Correct Answer: A. It can focus computation on relevant areas and reduce unnecessary processing

Why this is correct: It can focus computation on relevant areas and reduce unnecessary processing directly matches the concept tested.

Why B is incorrect: It always improves model accuracy does not correctly answer the question.

Why C is incorrect: It removes the need for labels does not correctly answer the question.

Why D is incorrect: It increases every image's resolution does not correctly answer the question.

Question 13: What is non-maximum suppression based on?

A. Comparing detection scores and overlap to suppress redundant detections

B. Comparing cloud bills

C. Comparing CPU temperatures

D. Comparing file names

Correct Answer: A. Comparing detection scores and overlap to suppress redundant detections

Why this is correct: Comparing detection scores and overlap to suppress redundant detections directly matches the concept tested.

Why B is incorrect: Comparing cloud bills does not correctly answer the question.

Why C is incorrect: Comparing CPU temperatures does not correctly answer the question.

Why D is incorrect: Comparing file names does not correctly answer the question.

Question 14: Why is IoU important when evaluating a detected object?

A. It quantifies how well a predicted region overlaps the reference region

B. It measures OCR character count

C. It measures API latency

D. It measures RAM

Correct Answer: A. It quantifies how well a predicted region overlaps the reference region

Why this is correct: It quantifies how well a predicted region overlaps the reference region directly matches the concept tested.

Why B is incorrect: It measures OCR character count does not correctly answer the question.

Why C is incorrect: It measures API latency does not correctly answer the question.

Why D is incorrect: It measures RAM does not correctly answer the question.

Question 15: What is a segmentation ground-truth mask?

A. A reference pixel-level annotation used to evaluate a segmentation prediction

B. A cloud access key

C. A camera setting

D. A Python package

Correct Answer: A. A reference pixel-level annotation used to evaluate a segmentation prediction

Why this is correct: A reference pixel-level annotation used to evaluate a segmentation prediction directly matches the concept tested.

Why B is incorrect: A cloud access key does not correctly answer the question.

Why C is incorrect: A camera setting does not correctly answer the question.

Why D is incorrect: A Python package does not correctly answer the question.

Question 16: What is OCR post-processing?

A. Processing extracted text or structured OCR output to improve usability or consistency

B. Changing the camera sensor

C. Training the GPU

D. Changing a model's architecture

Correct Answer: A. Processing extracted text or structured OCR output to improve usability or consistency

Why this is correct: Processing extracted text or structured OCR output to improve usability or consistency directly matches the concept tested.

Why B is incorrect: Changing the camera sensor does not correctly answer the question.

Why C is incorrect: Training the GPU does not correctly answer the question.

Why D is incorrect: Changing a model's architecture does not correctly answer the question.

Question 17: Why can OCR accuracy vary across documents?

A. Font, layout, image quality, language, orientation, and other factors can affect recognition

B. OCR results are always identical

C. Documents contain no visual variation

D. Cloud storage determines character shapes

Correct Answer: A. Font, layout, image quality, language, orientation, and other factors can affect recognition

Why this is correct: Font, layout, image quality, language, orientation, and other factors can affect recognition directly matches the concept tested.

Why B is incorrect: OCR results are always identical does not correctly answer the question.

Why C is incorrect: Documents contain no visual variation does not correctly answer the question.

Why D is incorrect: Cloud storage determines character shapes does not correctly answer the question.

Question 18: What is object tracking useful for in video?

A. Maintaining an object's identity or location across successive frames

B. Converting text into speech

C. Creating a database

D. Changing color spaces

Correct Answer: A. Maintaining an object's identity or location across successive frames

Why this is correct: Maintaining an object's identity or location across successive frames directly matches the concept tested.

Why B is incorrect: Converting text into speech does not correctly answer the question.

Why C is incorrect: Creating a database does not correctly answer the question.

Why D is incorrect: Changing color spaces does not correctly answer the question.

Question 19: Why can edge AI be useful for camera applications?

A. It can process visual data close to the camera and reduce dependence on round trips to the cloud

B. It eliminates the need for models

C. It always has unlimited compute

D. It prevents any network use

Correct Answer: A. It can process visual data close to the camera and reduce dependence on round trips to the cloud

Why this is correct: It can process visual data close to the camera and reduce dependence on round trips to the cloud directly matches the concept tested.

Why B is incorrect: It eliminates the need for models does not correctly answer the question.

Why C is incorrect: It always has unlimited compute does not correctly answer the question.

Why D is incorrect: It prevents any network use does not correctly answer the question.

Question 20: Why can annotation consistency matter in supervised computer vision?

A. Inconsistent labels can make the learning target ambiguous and reduce model reliability

B. It changes camera hardware

C. It removes the need for validation

D. It increases cloud storage automatically

Correct Answer: A. Inconsistent labels can make the learning target ambiguous and reduce model reliability

Why this is correct: Inconsistent labels can make the learning target ambiguous and reduce model reliability directly matches the concept tested.

Why B is incorrect: It changes camera hardware does not correctly answer the question.

Why C is incorrect: It removes the need for validation does not correctly answer the question.

Why D is incorrect: It increases cloud storage automatically does not correctly answer the question.

Phase 5 — Huawei Cloud + DevOps/MLOps

20 questions in this phase.

Question 1: What is a ModelArts dataset in an AI workflow?

A. A managed collection of data used for model development or related AI tasks

B. A physical camera

C. A network cable

D. A Python keyword

Correct Answer: A. A managed collection of data used for model development or related AI tasks

Why this is correct: A managed collection of data used for model development or related AI tasks directly matches the concept tested.

Why B is incorrect: A physical camera does not correctly answer the question.

Why C is incorrect: A network cable does not correctly answer the question.

Why D is incorrect: A Python keyword does not correctly answer the question.

Question 2: Why is dataset organization important in ModelArts?

A. Well-organized data supports repeatable labeling, training, and evaluation workflows

B. It changes CPU speed

C. It replaces models

D. It eliminates testing

Correct Answer: A. Well-organized data supports repeatable labeling, training, and evaluation workflows

Why this is correct: Well-organized data supports repeatable labeling, training, and evaluation workflows directly matches the concept tested.

Why B is incorrect: It changes CPU speed does not correctly answer the question.

Why C is incorrect: It replaces models does not correctly answer the question.

Why D is incorrect: It eliminates testing does not correctly answer the question.

Question 3: What is AutoML experiment management intended to support?

A. Running and comparing model-building experiments under different configurations

B. Editing documents

C. Managing classroom hardware

D. Changing image file extensions

Correct Answer: A. Running and comparing model-building experiments under different configurations

Why this is correct: Running and comparing model-building experiments under different configurations directly matches the concept tested.

Why B is incorrect: Editing documents does not correctly answer the question.

Why C is incorrect: Managing classroom hardware does not correctly answer the question.

Why D is incorrect: Changing image file extensions does not correctly answer the question.

Question 4: What is an inference endpoint client?

A. Software that sends properly formatted requests to a deployed model service

B. A training dataset

C. A camera lens

D. A storage bucket

Correct Answer: A. Software that sends properly formatted requests to a deployed model service

Why this is correct: Software that sends properly formatted requests to a deployed model service directly matches the concept tested.

Why B is incorrect: A training dataset does not correctly answer the question.

Why C is incorrect: A camera lens does not correctly answer the question.

Why D is incorrect: A storage bucket does not correctly answer the question.

Question 5: Why should a cloud API client check response status?

A. A request can fail even when the client successfully connects to the service

B. Status codes are only for images

C. They change model weights

D. They create training labels

Correct Answer: A. A request can fail even when the client successfully connects to the service

Why this is correct: A request can fail even when the client successfully connects to the service directly matches the concept tested.

Why B is incorrect: Status codes are only for images does not correctly answer the question.

Why C is incorrect: They change model weights does not correctly answer the question.

Why D is incorrect: They create training labels does not correctly answer the question.

Question 6: What is a timeout in an API call?

A. A condition where a response is not received within the allowed time

B. A model accuracy metric

C. A dataset label

D. A camera setting

Correct Answer: A. A condition where a response is not received within the allowed time

Why this is correct: A condition where a response is not received within the allowed time directly matches the concept tested.

Why B is incorrect: A model accuracy metric does not correctly answer the question.

Why C is incorrect: A dataset label does not correctly answer the question.

Why D is incorrect: A camera setting does not correctly answer the question.

Question 7: Why should cloud credentials not be hard-coded into public source code?

A. Exposed credentials can be abused to access protected resources

B. Hard-coding always lowers accuracy

C. Credentials affect image colors

D. It changes the model architecture

Correct Answer: A. Exposed credentials can be abused to access protected resources

Why this is correct: Exposed credentials can be abused to access protected resources directly matches the concept tested.

Why B is incorrect: Hard-coding always lowers accuracy does not correctly answer the question.

Why C is incorrect: Credentials affect image colors does not correctly answer the question.

Why D is incorrect: It changes the model architecture does not correctly answer the question.

Question 8: What is a CI build artifact?

A. A file or package produced by an automated build process

B. A camera frame

C. A model label

D. A cloud password

Correct Answer: A. A file or package produced by an automated build process

Why this is correct: A file or package produced by an automated build process directly matches the concept tested.

Why B is incorrect: A camera frame does not correctly answer the question.

Why C is incorrect: A model label does not correctly answer the question.

Why D is incorrect: A cloud password does not correctly answer the question.

Question 9: Why can automated CI tests run before deployment?

A. They can catch defects before a change reaches a deployment environment

B. They guarantee no production failures

C. They replace monitoring

D. They train every model

Correct Answer: A. They can catch defects before a change reaches a deployment environment

Why this is correct: They can catch defects before a change reaches a deployment environment directly matches the concept tested.

Why B is incorrect: They guarantee no production failures does not correctly answer the question.

Why C is incorrect: They replace monitoring does not correctly answer the question.

Why D is incorrect: They train every model does not correctly answer the question.

Question 10: What is a deployment pipeline?

A. An automated sequence of steps that moves validated software or models toward a target environment

B. A camera sensor

C. A spreadsheet

D. A single Python statement

Correct Answer: A. An automated sequence of steps that moves validated software or models toward a target environment

Why this is correct: An automated sequence of steps that moves validated software or models toward a target environment directly matches the concept tested.

Why B is incorrect: A camera sensor does not correctly answer the question.

Why C is incorrect: A spreadsheet does not correctly answer the question.

Why D is incorrect: A single Python statement does not correctly answer the question.

Question 11: Why is environment consistency important in AI deployment?

A. Differences in dependencies or runtime configuration can cause unexpected behavior

B. It only affects monitor brightness

C. It changes labels

D. It eliminates testing

Correct Answer: A. Differences in dependencies or runtime configuration can cause unexpected behavior

Why this is correct: Differences in dependencies or runtime configuration can cause unexpected behavior directly matches the concept tested.

Why B is incorrect: It only affects monitor brightness does not correctly answer the question.

Why C is incorrect: It changes labels does not correctly answer the question.

Why D is incorrect: It eliminates testing does not correctly answer the question.

Question 12: What is a model registry conceptually used for?

A. Keeping track of model versions and their lifecycle status

B. Storing keyboard settings

C. Creating OCR text

D. Changing camera exposure

Correct Answer: A. Keeping track of model versions and their lifecycle status

Why this is correct: Keeping track of model versions and their lifecycle status directly matches the concept tested.

Why B is incorrect: Storing keyboard settings does not correctly answer the question.

Why C is incorrect: Creating OCR text does not correctly answer the question.

Why D is incorrect: Changing camera exposure does not correctly answer the question.

Question 13: Why should production models have identifiable versions?

A. Teams need to know exactly which model is serving predictions

B. Versions change image colors

C. It removes the need for logs

D. It guarantees accuracy

Correct Answer: A. Teams need to know exactly which model is serving predictions

Why this is correct: Teams need to know exactly which model is serving predictions directly matches the concept tested.

Why B is incorrect: Versions change image colors does not correctly answer the question.

Why C is incorrect: It removes the need for logs does not correctly answer the question.

Why D is incorrect: It guarantees accuracy does not correctly answer the question.

Question 14: What is a rollback strategy?

A. A planned method for restoring a previously trusted deployment

B. A method for increasing image size

C. A data-labeling method

D. An OCR algorithm

Correct Answer: A. A planned method for restoring a previously trusted deployment

Why this is correct: A planned method for restoring a previously trusted deployment directly matches the concept tested.

Why B is incorrect: A method for increasing image size does not correctly answer the question.

Why C is incorrect: A data-labeling method does not correctly answer the question.

Why D is incorrect: An OCR algorithm does not correctly answer the question.

Question 15: What is observability in an AI service?

A. The ability to understand system behavior using metrics, logs, traces, and related signals

B. A method for resizing images

C. A type of neural layer

D. A storage format

Correct Answer: A. The ability to understand system behavior using metrics, logs, traces, and related signals

Why this is correct: The ability to understand system behavior using metrics, logs, traces, and related signals directly matches the concept tested.

Why B is incorrect: A method for resizing images does not correctly answer the question.

Why C is incorrect: A type of neural layer does not correctly answer the question.

Why D is incorrect: A storage format does not correctly answer the question.

Question 16: Why monitor API failure rates?

A. A rising failure rate can indicate availability, integration, authentication, or infrastructure problems

B. It changes the model labels

C. It improves OCR automatically

D. It increases dataset size

Correct Answer: A. A rising failure rate can indicate availability, integration, authentication, or infrastructure problems

Why this is correct: A rising failure rate can indicate availability, integration, authentication, or infrastructure problems directly matches the concept tested.

Why B is incorrect: It changes the model labels does not correctly answer the question.

Why C is incorrect: It improves OCR automatically does not correctly answer the question.

Why D is incorrect: It increases dataset size does not correctly answer the question.

Question 17: What is the difference between data drift and concept drift?

A. Data drift concerns changes in input distributions, while concept drift concerns changes in the input-target relationship

B. They are identical hardware failures

C. Both refer only to monitor changes

D. Neither can affect ML systems

Correct Answer: A. Data drift concerns changes in input distributions, while concept drift concerns changes in the input-target relationship

Why this is correct: Data drift concerns changes in input distributions, while concept drift concerns changes in the input-target relationship directly matches the concept tested.

Why B is incorrect: They are identical hardware failures does not correctly answer the question.

Why C is incorrect: Both refer only to monitor changes does not correctly answer the question.

Why D is incorrect: Neither can affect ML systems does not correctly answer the question.

Question 18: Why should retraining pipelines include validation before deployment?

A. A newly trained model may be worse than the current production model

B. Validation only changes the UI

C. Every retrained model is automatically better

D. Deployment cannot be reversed

Correct Answer: A. A newly trained model may be worse than the current production model

Why this is correct: A newly trained model may be worse than the current production model directly matches the concept tested.

Why B is incorrect: Validation only changes the UI does not correctly answer the question.

Why C is incorrect: Every retrained model is automatically better does not correctly answer the question.

Why D is incorrect: Deployment cannot be reversed does not correctly answer the question.

Question 19: What is an MLOps audit trail useful for?

A. Tracing important changes and decisions across data, models, code, and deployments

B. Changing camera settings

C. Increasing storage speed

D. Creating random labels

Correct Answer: A. Tracing important changes and decisions across data, models, code, and deployments

Why this is correct: Tracing important changes and decisions across data, models, code, and deployments directly matches the concept tested.

Why B is incorrect: Changing camera settings does not correctly answer the question.

Why C is incorrect: Increasing storage speed does not correctly answer the question.

Why D is incorrect: Creating random labels does not correctly answer the question.

Question 20: Why does the course include a capstone project?

A. It provides an opportunity to combine AI development, deployment, testing, and presentation skills

B. It is only for hardware repair

C. It replaces every course topic

D. It avoids practical work

Correct Answer: A. It provides an opportunity to combine AI development, deployment, testing, and presentation skills

Why this is correct: It provides an opportunity to combine AI development, deployment, testing, and presentation skills directly matches the concept tested.

Why B is incorrect: It is only for hardware repair does not correctly answer the question.

Why C is incorrect: It replaces every course topic does not correctly answer the question.

Why D is incorrect: It avoids practical work does not correctly answer the question.

Final Answer Key — 4th Edition

Phase 1 — Artificial Intelligence (AI)

1. A 2. A 3. A 4. A 5. A 6. A 7. A 8. A 9. A 10. A
11. A 12. A 13. A 14. A 15. A 16. A 17. A 18. A 19. A 20. A

Phase 2 — Machine Learning (ML)

21. A 22. A 23. A 24. A 25. A 26. A 27. A 28. A 29. A 30. A
31. A 32. A 33. A 34. A 35. A 36. A 37. A 38. A 39. A 40. A

Phase 3 — Deep Learning (DL)

41. A 42. A 43. A 44. A 45. A 46. A 47. A 48. A 49. A 50. A
51. A 52. A 53. A 54. A 55. A 56. A 57. A 58. A 59. A 60. A

Phase 4 — Computer Vision (CV)

61. A 62. A 63. A 64. A 65. A 66. A 67. A 68. A 69. A 70. A
71. A 72. A 73. A 74. A 75. A 76. A 77. A 78. A 79. A 80. A

Phase 5 — Huawei Cloud + DevOps/MLOps

81. A 82. A 83. A 84. A 85. A 86. A 87. A 88. A 89. A 90. A
91. A 92. A 93. A 94. A 95. A 96. A 97. A 98. A 99. A 100. A